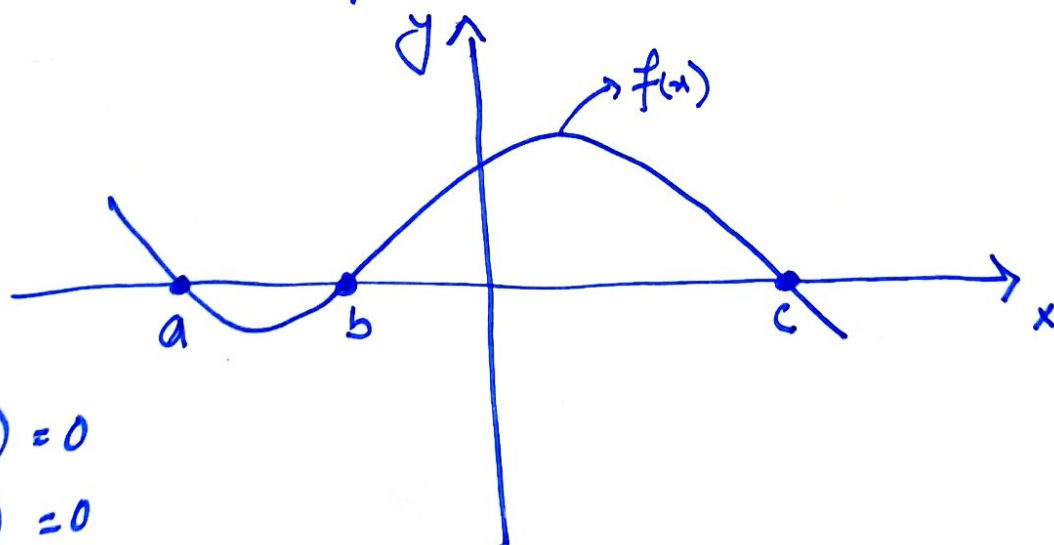


Lecture/Week 3

Roots of non-linear functionsZeros " " "Solution of non-linear equations.

For a single variable function $f(x)$, a point $x=a$ is called a root or a zero of $f(x)$ if $f(a) = 0$. That is, the value of function $f(x)$ is zero at $x=a$.



$$f(a) = 0$$

$$f(b) = 0$$

$$f(c) = 0$$

Therefore a, b, c are roots or f zeros of given function $f(x)$.

Example ①

$$f(x) = x^2 + 4x - 10$$

$$x^2 + 4x - 10 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{16 - 4(-10)}}{2(1)} = \frac{-4 \pm \sqrt{16+40}}{2}$$

$$x = \frac{-4 \pm \sqrt{56}}{2}$$

$$x = 1.74166$$

$$x = -5.74166 \quad \text{Ans.}$$

Example ②

$$f(x) = \cos 2x$$

Example ③

$$T(x) = x - \frac{1+3x^2}{4+x^3}$$

Example ④

$$G(x) = x^{1/3} - x^2 - 8$$

Numerical Methods

- 1) Bisection Method
- 2) Regula Falsi Method (Method of False Position)
- 3) Secant Method
- 4) Newton's Method

Bisection Method

(3)

The Bisection Method is based on the intermediate value theorem (Mean value theorem)

- i) If $f(x)$ is continuous on an interval $[a, b]$
- ii) $f(a)$ and $f(b)$ differ in sign, that is
 $f(a) \cdot f(b) < 0$.

Then there exist a point 'c' in the interval $[a, b]$
for which $f(x) = 0$
or $f(c) = 0$

'c' is a root.

In bisection method, we assume

$$c = \frac{a+b}{2}$$

midpoint of the interval $[a, b]$.

Example

Approximate the positive roots of $f(x) = x^2 + 4x - 10$ correct to 2-decimal places by using the Bisection Method.

Solution

$$f(x) = x^2 + 4x - 10$$

Step I :- Finding Interval $[a \ b]$

$$f(0) = -10$$

$$f(1) = -5$$

$$f(-1) = -13$$

$$f(2) = 2$$

$$[0 \ 2]$$

$$[1 \ 2]$$

$$[-1 \ 2]$$

Interval.

$$a = 1$$

$$b = 2$$

Step II ..

<u>n</u>	<u>a</u>	<u>f(a)</u>	<u>b</u>	<u>f(b)</u>	<u>$c = \frac{a+b}{2}$</u>	<u>f(c)</u>
1	1	$\boxed{-5}$	2	$\boxed{2}$	$c = \frac{1+2}{2} = 1.5$	$\boxed{f(1.5) = -1.75}$
2	1.5	$\boxed{-1.75}$	2	$\boxed{2}$	$c = \frac{1.5+2}{2} = 1.75$	$\boxed{f(1.75) = 0.0625}$
3	1.5	-1.75	1.75	0.0625	$c = \frac{1.5+1.75}{2} = 1.625$	-0.859375
4	1.625	-0.859375	1.75	0.0625	$c = 1.6875$	-0.402344
5	1.6875	-0.402344	1.75	0.0625	$c = 1.71875$	-0.170898
6	1.71875	-0.170898	1.75	0.0625	$c = 1.734375$	-0.054443
7	1.734375	-0.054443	1.75	0.0625	$\boxed{c = 1.742188}$	$\boxed{0.003971}$

↓
Root.

Example (2)

$$f(x) = e^{-x} (-1.35 \cos x - 1.85 \sin x)$$

Approximate the critical point of $f(x)$ using Bisection method with initial interval $[3 \ 4]$. Perform three iterations.

Solution For critical point $f'(x) = 0$

$$f'(x) = e^{-x} [3.25 \sin x - 0.5 \cos x]$$

To approximate zero of $f'(x)$, let

$$F(x) = e^{-x} [3.25 \sin x - 0.5 \cos x]$$

Given $a = 3$

$b = 4$

<u>n</u>	<u>a</u>	<u>f(a)</u>	<u>b</u>	<u>F(b)</u>	<u>$c = \frac{a+b}{2}$</u>	<u>F(c)</u>
1	3	<u>0.047127</u>	4	<u>-0.038372</u>	3.5	<u>-0.0202871</u>
2	3	0.047127	3.5	-0.0202871	3.25	0.00563896
3	3.25	0.00563896	3.5	-0.0202871	<u>3.375</u>	<u>-0.0090768</u>

Ans:

(6)

The Regula Falsi Method

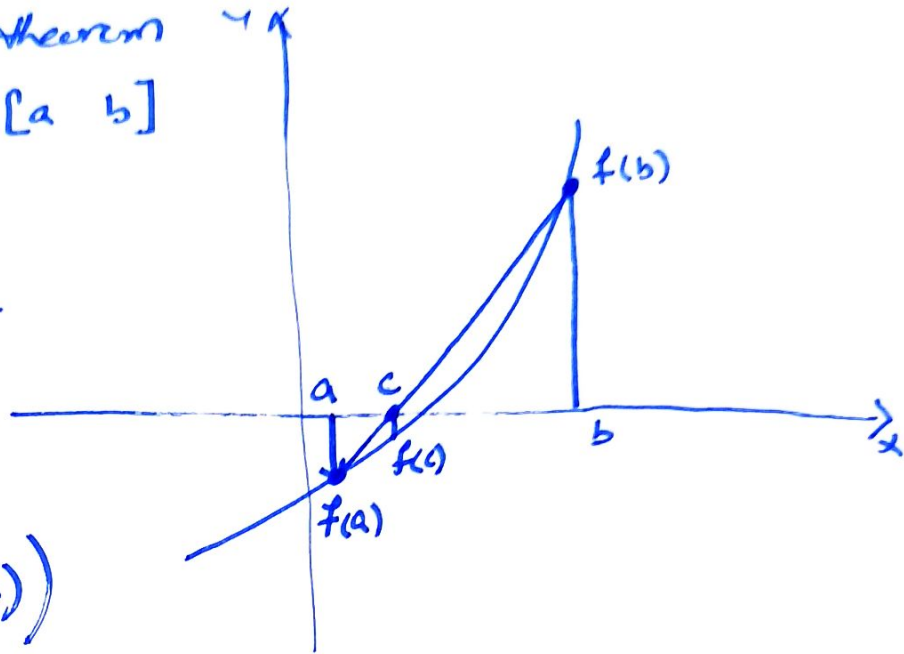
Method of False Position

1) Based on mean value theorem
continuous on interval $[a, b]$

$$f(a) * f(b) < 0$$

There exist 'c' such that

$$f(c) = 0$$



$$(a, f(a)) \quad (b, f(b))$$

Equation of line

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y - f(a) = \frac{f(b) - f(a)}{b - a} (x - a)$$

The point where line cuts the x-axis is

$$y = 0 \quad \text{and} \quad x = c$$

$$0 - f(a) = \frac{f(b) - f(a)}{b - a} (c - a)$$

$$c = a - \frac{b - a}{f(b) - f(a)} f(a)$$

Formula

Example

$$f(x) = x^2 + 4x - 10$$

Solution

$$f(0) = -10 \quad f(1) = -5$$

$$f(-1) = -13 \quad f(2) = 2$$

$$[0 \quad 2] \quad \underbrace{[1 \quad 2]}_{a=1 \quad b=2} \quad [-1 \quad 2]$$

$$c = a - \frac{b-a}{f(b)-f(a)} f(a)$$

<u>n</u>	<u>a</u>	<u>f(a)</u>	<u>b</u>	<u>f(b)</u>	<u>c</u>	<u>f(c)</u>
1	1	-5	2	2	$c = 1 - \frac{(2-1)}{(2-(-5))} (-5)$ $= 1.714286$	$f(1.714286)$ $= -0.204080$
2	1.714286	-0.204080	2	2	1.740741	-0.006857
3	1.740741	-0.006857	2	2	1.741627	-0.000227
4	1.741627	-0.000227	2	2	1.741656	-0.000010

Root.